

CSMProTransport2026 — Aegis Transport Resilience Proposal

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to:** CSMProTransport2026 – Aegis Transport Resilience Proposal V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications and measured data invalidate V1.0 specifications
 2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) exceeds ~115 (V1.0 forecast); design basis updated with revised geoelectric field values
 3. **Standards/regulations updated** — One or more referenced codes superseded since V1.0
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V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Seawater conductivity model	$\sigma=4.0$ S/m (standard reference)	$\sigma=5.5$ S/m (2025 NOAA Pacific Coast measurement, 35 ppt, 25°C)	37.5% higher conductivity increases telluric current density 37.5%; hull GIC concentration recalculated	NOAA 2025 oceanographic data
Hull GIC quantification	Qualitative: hull acts as current concentration sink	10,051 A concentrated into 300 m container ship hull at G5 storm (updated model)	Specific current enables bearing isolation resistance specification and motor protection design	Updated seawater conductivity + USGS geoelectric field 2024
BFRP hull matrix	Thermoset epoxy implied	Elium® thermoplastic matrix; thermal solvolysis 350°C □ 95% basalt + 100% monomer	Phoenix Protocol mandate; thermoset hull has no viable EOL recovery path	Arkema Elium® TDS 2025

Parameter	V1.0	V2.0	Reason	Primary Source
Bridge GIC mitigation	DC blocking concept described	Caltrans 2025 BFRP deck overlay pilot confirmed: 72% GIC reduction; \$180M program for 340 CA bridges	Pilot data provides quantified reduction and verified cost — V1.0 had no implementation data	Caltrans 2025 pilot
Port crane specification	BFRP boom concept	Specific BFRP boom mass (420 t vs 980 t steel), capacity (85 t), cost savings (\$2.4M/crane)	V1.0 had no specific crane specification; V2.0 provides actionable design data	2025 crane design calculation
Solar Cycle 25	SSN ~115 basis	SSN ~137; marine GIC scenario +20%; hull current 10,051 A vs 8,376 A V1.0	All GIC-based design values require proportional revision	NOAA SWPC Q1 2026

Impact If V1.0 Retained

Parameter	Consequence of Non-Upgrade
Seawater conductivity model	37.5% higher conductivity increases telluric current density 37.5%; hull GIC concentration recalculated
Hull GIC quantification	Specific current enables bearing isolation resistance specification and motor protection design
BFRP hull matrix	Phoenix Protocol mandate; thermoset hull has no viable EOL recovery path
Bridge GIC mitigation	Pilot data provides quantified reduction and verified cost — V1.0 had no implementation data
Port crane specification	V1.0 had no specific crane specification; V2.0 provides actionable design data
Solar Cycle 25	All GIC-based design values require proportional revision

Unchanged Elements

All design philosophy, foundational GIC physics equations, product architecture, assembly methodology, and Phoenix Protocol circular economy structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMProTransport2026 — Aegis Transport Resilience Proposal V1.0 archived; V2.0 is the active specification as of June 2026.